Taylor dispersion in regular polygonal ducts: the sign of the skewness at all times

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Abstract. Aminian, Camassa and McLaughlin claimed from finite-element computations that among regular polygonal ducts the equilateral triangle is the only one whose solute skewness changes sign, "all other polygons possess positive skewness at all times", adding that this "cannot be proved on finite times". We prove the claim. For every regular n-gon with $n \geq 4$, every Péclet number and every time, the skewness of a cross-sectionally uniform release with symmetric longitudinal profile is strictly positive; the same holds for the disk, and for the equilateral triangle the skewness changes sign exactly once. The proof has four parts. An identity writes the third moment as $K(\tau) = 2\int_0^\tau \langle u\, A(s) A(\tau - s)\rangle \, \mathrm{d}s$, where u is the mean-free Poiseuille profile and A is the time integral of the Neumann heat flow of u. A maximum-principle argument, valid on every convex section, gives an explicit lower bound for K that is positive on $0 < \tau \leq 0.015$ for all n at once, even though $\langle u^3\rangle \to 0$ as $n \to \infty$. A modal approximation of A with an a-posteriori energy estimate certifies $K > 0$ on $[0.015, \infty)$ in ball arithmetic for $4 \leq n \leq 199$. For $n \geq 200$ the same estimate is applied with Bessel modes and closed-form disk data, with errors bounded uniformly in n. The paper also collects the long- and short-time results of the same problem.

The question

M. Aminian, R. Camassa and R. M. McLaughlin [1] [2] study a passive tracer released uniformly across a straight duct in pressure-driven flow. The sign of the skewness of its cross-sectionally averaged distribution says whether the tracer is front- or back-loaded. The abstract of the arXiv version (v1–v3) says:

> *"Interestingly, the equilateral triangle is the only regular polygon with this property, all other polygons possess positive skewness at all times, although this cannot cannot be proved on finite times due to the lack of closed form flow solutions for such geometries."*

"This property" means positive short-time and negative long-time skewness. The published abstract weakens the sentence to "The equilateral triangle appears to be the only regular polygon with this property—all others possess positive skewness at all times." The evidence in the source is finite-element computation of the two asymptotic coefficients for finitely many n. An earlier stage of the present work [3] settled the two asymptotic signs (Theorem 4). The sign at intermediate times was open.

Let Ω be the section, $\tilde{u}$ the Poiseuille profile ($-\Delta\tilde{u} = 2$, $\tilde{u} = 0$ on $\partial\Omega$), $u = \tilde{u} - \langle\tilde{u}\rangle$, and $P_s = \mathrm{e}^{s\Delta}$ the Neumann heat semigroup on Ω. Put

$$A(s) = \int_0^s P_r u \, \mathrm{d}r, \qquad K(\tau) = 2\int_0^\tau \langle u\, A(s)\, A(\tau - s)\rangle \, \mathrm{d}s. \tag{1}$$

By Proposition 6, the third central moment of the release is $\mu_3(\tau) = \kappa_3 + 3\mathrm{Pe}^3 K(\tau)$, where κ_3 is the third cumulant of the initial longitudinal profile, and $\mu_2(\tau) > 0$. So for a symmetric longitudinal profile, in particular a plane source, the sign of the skewness is the sign of $K(\tau)$ for every Péclet number.

Theorem 2. For every integer $n \geq 4$ and the regular n-gon, $K(\tau) > 0$ for all $\tau > 0$. Hence the skewness is strictly positive at all times, for every Péclet number and every cross-sectionally uniform release with symmetric longitudinal profile. The same holds for the disk.

Theorem 3. For the equilateral triangle K has exactly one zero τ^* on $(0, \infty)$, simple, with $K > 0$ before and $K < 0$ after; $\tau^* \in [1.17, 1.25]$ in the time unit a^2/κ, a the inradius.

Theorem 4 (the two ends)**.** Let g be the mean-free solution of $-\Delta g = u$, $\partial_n g = 0$. Then $K(\tau) = 2\langle ug^2\rangle\tau + O(1)$ as $\tau \to \infty$ and $K(\tau) = \langle u^3\rangle\tau^3/3 + O(\tau^{7/2})$ as $\tau \to 0$, and

(1) $\langle ug^2\rangle = -7101/476476000 < 0$ for $n = 3$, and $\langle ug^2\rangle > 0$ for every $n \geq 4$ and for the disk;

(2) $\langle u^3\rangle = 27/19250$ for $n = 3$ (inradius 1), $\langle u^3\rangle > 0$ for every $n \geq 4$, $\langle u^3\rangle = 0$ for the disk, and $\langle u^3\rangle = \tan^4(\pi/n)/360 + O(\tan^5(\pi/n))$.

In the notation of the source $2\langle ug^2\rangle = \langle ug_2\rangle$, see (8). The certified quantities behind Theorem 2 reprove item 1 of Theorem 4 independently of [3]. Section 8 gives enclosures of all entries of Table 1 of the source.

Remark 5. Theorem 2 needs $\kappa_3 = 0$. For a non-symmetric longitudinal profile $\mu_3 = \kappa_3 + 3\mathrm{Pe}^3 K$, and the sign depends on the data. The results concern the moments of the Aris hierarchy, exactly as in the source. ⌟

1 Exact formulas

Throughout, P_n is the regular n-gon with apothem 1 and side $\{x = 1,\ |y| \leq t\}$, $t = \tan(\pi/n)$, $R = \sec(\pi/n) = \sqrt{1+t^2}$, $|P_n| = nt$, $|\partial P_n| = 2nt$. For a section Ω we write $\langle f\rangle = |\Omega|^{-1}\int_\Omega f$, $\|f\|^2 = \langle f^2\rangle$, $\|f\|_\partial^2 = |\Omega|^{-1}\oint_{\partial\Omega} f^2$. Further $c = \langle\tilde{u}\rangle$, $u = \tilde{u} - c$, $m = -\partial_n\tilde{u} \geq 0$ (with $|\Omega|^{-1}\oint m = 2$). All statements are invariant under dilations, with τ scaling like a length squared. We write g for the function g_1 of the source and $\mathcal{T}(f,h) = \langle ufh\rangle$.

The tracer solves $T_\tau + \mathrm{Pe}\,uT_x = \Delta_3 T$ in $\mathbb{R}\times\Omega$, $\partial_n T = 0$, $T(x,\cdot,0) = \varrho(x)$, $\varrho \geq 0$, $\int\varrho = 1$, with finite moments, and $\int x\varrho = 0$ after a translation. The moments $c_p(\cdot,\tau) = \int x^p T\,\mathrm{d}x$ solve the Aris hierarchy $\partial_\tau c_p = \Delta c_p + p(p-1)c_{p-2} + p\,\mathrm{Pe}\,uc_{p-1}$, $\partial_n c_p = 0$, $c_p(0) = \int x^p\varrho$ [4].

Proposition 6. $\mu_2(\tau) > 0$ and $\mu_3(\tau) = \kappa_3 + 3\mathrm{Pe}^3 K(\tau)$ with K as in (1). Moreover, with $G(a,b) = \langle u\,P_a u\,P_b u\rangle$,

$$K(\tau) = 2\iint_{a,b\geq 0,\ a+b\leq\tau} (\tau - a - b)\,G(a,b)\,\mathrm{d}a\,\mathrm{d}b, \tag{7}$$

$A(s) = g - P_s g$, and $K(\tau) = 2\langle ug^2\rangle\tau + O(1)$ as $\tau \to \infty$. If g_2 is the mean-free

solution of $-\Delta g_2 = 2(ug - \langle ug\rangle)$, $\partial_n g_2 = 0$, then $\langle ug_2\rangle = 2\langle ug^2\rangle$ and

$$\tau^{1/2}\mathrm{Sk}(\tau) \to \frac{3\mathrm{Pe}^3\langle ug_2\rangle}{(2+2\mathrm{Pe}^2\langle ug\rangle)^{3/2}}\ , \tag{8}$$

which is equation (17) of the source.

Proof. $c_0 = 1$, $c_1 = \mathrm{Pe}\,A$, $\langle c_1\rangle = 0$. Next $c_2 = \int x^2\varrho + 2\tau + \mathrm{Pe}^2 S$ with $S' = \Delta S + 2uA$, $S(0) = 0$. Hence $\mu_2' = 2 + 2\mathrm{Pe}^2\langle uA\rangle$ with $\langle uA(s)\rangle = \int_0^s \|P_{r/2}u\|^2\,\mathrm{d}r \ge 0$, so $\mu_2 > 0$. Further $\frac{\mathrm{d}}{\mathrm{d}\tau}\langle c_3\rangle = 6\langle c_1\rangle + 3\mathrm{Pe}\langle uc_2\rangle = 3\mathrm{Pe}^3\langle uS\rangle$. By Duhamel and self-adjointness, $\langle uS(\sigma)\rangle = 2\int_0^\sigma \langle A'(\sigma - s)\,u\,A(s)\rangle\,\mathrm{d}s$. Integrating over $\sigma \in [s,\tau]$ with $A(0) = 0$ gives (1), and (7) follows by inserting $A(s) = \int_0^s P_r u\,\mathrm{d}r$: for $r + q \le \tau$ the variable s runs over an interval of length $\tau - r - q$. Spectrally $A(s) = g - P_s g$. Insert this into (1). The terms containing $P_s g$ stay bounded because of the spectral gap: $\int_0^\tau |\mathcal{T}(g, P_s g)|\,\mathrm{d}s$ is bounded, and $|\mathcal{T}(P_s g, P_{\tau-s}g)| \le C\mathrm{e}^{-\nu\tau}$. So $K = 2\langle ug^2\rangle\tau + O(1)$. For small τ, $\|P_a u - u\| \le 0.64\sqrt{a}\|\nabla u\|$ spectrally and u is bounded, so $|G(a,b) - \langle u^3\rangle| \le C(\sqrt{a} + \sqrt{b})$ and (7) gives $K = \langle u^3\rangle\tau^3/3 + O(\tau^{7/2})$. Testing the two weak forms against g_2 and g gives $\langle ug_2\rangle = \langle \nabla g \cdot \nabla g_2\rangle = 2\langle ug^2\rangle$; and $\mu_2 \sim (2 + 2\mathrm{Pe}^2\langle ug\rangle)\tau$. □

2 Tools

Let V be the space of D_n-invariant, mean-free functions in $H^1(P_n)$. Such a function is determined by its restriction to the half-sector $T = \{0 \le y \le tx,\ x \le 1\}$, a right triangle of diameter R, and its restriction is mean-free on T. The functions u, g, $P_s u$, $P_s g$, $A(s)$ lie in V. Payne–Weinberger on the convex set T [5] [6] gives

$$\|\varphi\| \le C_P\|\nabla\varphi\|, \quad C_P = \frac{R}{\pi}, \qquad \|P_s\varphi\| \le \mathrm{e}^{-\nu s}\|\varphi\|, \quad \nu = \frac{\pi^2}{R^2} \qquad (\varphi \in V). \tag{9}$$

Since $x \cdot n = 1$ on ∂P_n, $\|\varphi\|_\partial^2 = \langle \mathrm{div}(x\varphi^2)\rangle = 2\|\varphi\|^2 + 2\langle \varphi\, x \cdot \nabla\varphi\rangle$, so

$$\|\varphi\|_\partial \le \tau_{\mathrm{tr}}\|\nabla\varphi\|, \qquad \tau_{\mathrm{tr}}^2 = 2C_P^2 + 2RC_P. \tag{10}$$

For $R_* \in L^2$, $b \in L^2(\partial P_n)$ the functional $\ell(\varphi) = \langle R_*\varphi\rangle + |P_n|^{-1}\oint b\varphi$ on V has dual norm $\|\ell\|_* = \sup_{\varphi\in V}|\ell(\varphi)|/\|\nabla\varphi\| \le C_P\|R_* - \langle R_*\rangle\| + \tau_{\mathrm{tr}}\|b\|_\partial$.

Lemma 11. Let $E \in H^2(P_n)$ be harmonic. Then $\|E\|^2 \le \frac{1+R}{2}\|E\|_\partial^2$ and $\|\partial_n E\|_\partial \le (t + R)\|\partial_s E\|_\partial$.

Proof. $\tilde{u} \in H^2(P_n)$ by convexity [7]. Green's identities give $2\int E^2 = -2\int \tilde{u}|\nabla E|^2 + \oint E^2 m$. On a side, $\tilde{u} \le (1 - x)(x + R)$ by the maximum principle, so $m \le 1 + R$. The second bound is the Rellich identity: the field $(x \cdot \nabla E)\nabla E - \frac{1}{2}|\nabla E|^2 x$ is divergence-free. With $x \cdot n = 1$ and $x \cdot \nabla E = \partial_n E + \sigma\partial_s E$, $|\sigma| \le t$, on a side, this gives $X^2 \le Y^2 + 2tXY$ for $X = \|\partial_n E\|_\partial$, $Y = \|\partial_s E\|_\partial$. □

Lemma 12 (cell problem). Let $\tilde{u}_H$ be a D_n-invariant polynomial with $-\Delta\tilde{u}_H = 2$, and g_H a D_n-invariant polynomial with $-\Delta g_H = \tilde{u}_H - \langle\tilde{u}_H\rangle$. Then

$$\|g - (g_H - \langle g_H\rangle)\| \le C_P\Big(C_P\big(\tfrac{1+R}{2}\big)^{1/2}\|\tilde{u}_H\|_\partial + \tau_{\mathrm{tr}}\|\partial_n g_H\|_\partial\Big).$$

Proof. $E=\tilde u-\tilde u_H$ is harmonic, in H^2, and equals $-\tilde u_H$ on ∂P_n. For $w=g-g_H+\langle g_H\rangle\in V$ and $\varphi\in V$, $\langle\nabla w\cdot\nabla\varphi\rangle=\langle(E-\langle E\rangle)\varphi\rangle-|P_n|^{-1}\oint\varphi\,\partial_n g_H$. Use (9), (10) and Lemma 11. □

Lemma 13 (modal approximation of A). Let $\hat g\in V$, $\psi_1,\dots,\psi_M\in V$, $\mu_k>0$, $d_k\in\mathbb{R}$, and let $\ell_k(\varphi)=\langle\nabla\psi_k\cdot\nabla\varphi\rangle-\mu_k\langle\psi_k\varphi\rangle$ on V. Put $\tilde A(s)=\hat g-\sum_k d_k\mathrm{e}^{-\mu_k s}\psi_k$. If $\|g-\hat g\|\le\omega$, $\|\hat g-\sum_k d_k\psi_k\|\le\rho$ and $L_1=\sum_k|d_k|\|\ell_k\|_*(2\mu_k)^{-1/2}$, then

$$\|A(s)-\tilde A(s)\|\le\Delta_0:=2\omega+\rho+L_1\qquad(s\ge0).$$

Proof. $y(s)=P_sg-\sum_k d_k\mathrm{e}^{-\mu_k s}\psi_k$ satisfies $\langle y'\varphi\rangle+\langle\nabla y\cdot\nabla\varphi\rangle=-\sum_k d_k\mathrm{e}^{-\mu_k s}\ell_k(\varphi)$ on V. With $\varphi=y$, $\frac{\mathrm d}{\mathrm ds}\|y\|^2\le-2\|\nabla y\|^2+2\ell(s)\|\nabla y\|\le\ell(s)^2$, where $\ell(s)=\sum_k|d_k|\mathrm{e}^{-\mu_k s}\|\ell_k\|_*$. By Minkowski's inequality $\int_0^\infty\ell^2\le L_1^2$, so $\|y(s)\|\le\|y(0)\|+L_1\le\omega+\rho+L_1$. Finally $A(s)-\tilde A(s)=(g-\hat g)-y(s)$. □

Lemma 14 (comparison). In the setting of Lemma 13 let w be bounded with $\|w\|_\infty\le U$, $\|u-w\|_\infty\le o$, and $\tilde K(\tau)=2\int_0^\tau\langle w\tilde A(s)\tilde A(\tau-s)\rangle\,\mathrm ds$. Then $|K(\tau)-\tilde K(\tau)|\le\mathcal E(\tau)$, where

$$\mathcal E(\tau)=2U\bigl(2\Delta_0\min(\tfrac12\tau^2\|u\|,\tau\|g\|)+\Delta_0^2\tau\bigr)+2o\min(\tfrac16\tau^3\|u\|^2,\tau\|g\|^2).$$

Proof. $\|A(s)\|\le\min(s\|u\|,\|g\|)$ by the spectral theorem. With $e=A-\tilde A$, $\langle uA(s)A(s')\rangle-\langle w\tilde A(s)\tilde A(s')\rangle=\langle(u-w)A(s)A(s')\rangle+\langle we(s)A(s')\rangle+\langle w\tilde A(s)e(s')\rangle$ and $\|\tilde A\|\le\|A\|+\Delta_0$. Integrate. □

If $\tilde A(s)=\sum_{k=0}^M D_k\mathrm{e}^{-\mu_k s}\psi_k$ with $\mu_0=0$, $\psi_0=\hat g$, $D_0=1$, $D_k=-d_k$, and $W_{kl}=D_kD_l\langle w\psi_k\psi_l\rangle$, then

$$\tilde K(\tau)=2\sum_{k,l=0}^{M}W_{kl}E_{kl}(\tau),\qquad E_{kl}(\tau)=\int_0^\tau\mathrm{e}^{-\mu_k a-\mu_l(\tau-a)}\,\mathrm da.\tag{15}$$

Writing $E_{kl}=\tau\int_0^1\mathrm{e}^{-\tau(\theta\mu_k+(1-\theta)\mu_l)}\,\mathrm d\theta$ and using $\sup_{x\ge0}(3x^2+x^3)\mathrm{e}^{-x}<3$ gives $|\partial_\tau^3E_{kl}|\le3\tau^{-2}$. So

$$|\tilde K'''(\tau)|\le6\tau^{-2}\sum_{k,l}|W_{kl}|,\tag{16}$$

and $0\le E_{kl}\le1/\max(\mu_k,\mu_l)$ for $(k,l)\ne(0,0)$, so that $\tilde K(\tau)\ge2W_{00}\tau-C_1$ with $C_1=2\sum_{(k,l)\ne(0,0)}|W_{kl}|/\max(\mu_k,\mu_l)$.

3 Short times, uniformly in the section

In this section Ω is any bounded convex polygon or the disk. Let $F(s)=\langle uP_su\rangle$. This is the only place where the boundary layer is treated, and no discretisation enters.

Lemma 17. $\Gamma_s:=P_su-u+2s\ge0$ and $\langle\Gamma_s\rangle=2s$. Moreover $\Lambda(s):=\langle\tilde u\,\Gamma_s\rangle$ satisfies

$$F(s)=\langle u^2\rangle-2cs+\Lambda(s),\qquad\Lambda(s)=\int_0^s|\Omega|^{-1}\oint_{\partial\Omega}m\,P_r\tilde u\,\mathrm d\sigma\,\mathrm dr.$$

Proof. For $\varphi \in H^1$ Green's identity gives $\langle \nabla u \cdot \nabla \varphi \rangle = 2\langle \varphi \rangle - |\Omega|^{-1} \oint m\varphi$. So $\langle \partial_s \Gamma_s \, \varphi \rangle + \langle \nabla \Gamma_s \cdot \nabla \varphi \rangle = |\Omega|^{-1} \oint m\varphi$, $\Gamma_0 = 0$. Since $m \geq 0$, testing with the negative part gives $\Gamma_s \geq 0$; testing with $\varphi = 1$ gives the mean. Next $\langle \tilde{u}\Gamma_s \rangle = \langle uP_s u \rangle + c\langle P_s u \rangle - \langle \tilde{u}u \rangle + 2sc = F(s) - \langle u^2 \rangle + 2cs$. Finally $F'(r) = -\langle \nabla u \cdot \nabla P_r u \rangle = |\Omega|^{-1} \oint m P_r u = |\Omega|^{-1} \oint m P_r \tilde{u} - 2c$. □

Lemma 18 (barrier)**.** Let d be the distance to $\partial\Omega$ and suppose $\tilde{u} \leq \bar{m}\, d + \beta$ on Ω. Then $\Lambda(s) \leq \frac{8}{3\sqrt{\pi}}\, \bar{m}\, s^{3/2} + 2\beta s$.

Proof. Let $Z(y, r) = \mathbb{E}|y + \sqrt{2r}\, N|$, N standard normal. It solves $Z_r = Z_{yy}$, with $Z(y, 0) = |y|$, $Z_y(0, r) = 0$, $Z_y \geq 0$ on $y \geq 0$, and $Z(0, r) = 2\sqrt{r/\pi}$. Put $\zeta(x, r) = Z(d(x), r)$. For $\varphi \geq 0$ in $H^1(\Omega)$ we get $\langle \zeta_r \varphi \rangle + \langle \nabla \zeta \cdot \nabla \varphi \rangle = \langle \nabla d \cdot \nabla (Z_y(d, r)\varphi) \rangle \geq 0$: the function $Z_y(d, r)\varphi$ is non-negative and vanishes on $\partial\Omega$, and d is concave. So ζ is a supersolution and $P_r d \leq \zeta$, in particular $P_r d \leq 2\sqrt{r/\pi}$ on $\partial\Omega$. Then $P_r \tilde{u} \leq \bar{m}\, 2\sqrt{r/\pi} + \beta$ on $\partial\Omega$. Insert this in Lemma 17, using $|\Omega|^{-1} \oint m = 2$. □

Proposition 19. With $E := c^2 - \langle u^2 \rangle$, for all $a, b \geq 0$,

$$G(a, b) \geq \langle u^3 \rangle + 2E(a + b) + 4abc - (c + 2b)\Lambda(a) - (c + 2a)\Lambda(b) - c\Lambda(a + b).$$

Proof. $G(a, b) = \langle \tilde{u} P_a u P_b u \rangle - cF(a + b)$. Write $P_a u = u - 2a + \Gamma_a$ and expand. Since $\tilde{u} \geq 0$ and $\Gamma \geq 0$: $\langle \tilde{u}\Gamma_a \Gamma_b \rangle \geq 0$; $\langle \tilde{u}(u - 2a)\Gamma_b \rangle = \langle \tilde{u}^2 \Gamma_b \rangle - (c + 2a)\Lambda(b) \geq -(c + 2a)\Lambda(b)$, and likewise with a, b exchanged; and $\langle \tilde{u}(u - 2a)(u - 2b) \rangle = \langle u^3 \rangle + c\langle u^2 \rangle - 2(a + b)\langle u^2 \rangle + 4abc$. Finally use $F(a + b) = \langle u^2 \rangle - 2c(a + b) + \Lambda(a + b)$. □

Proposition 20. Let $\langle u^3 \rangle \geq 0$, $\Lambda(s) \leq Ls^{3/2} + Bs$, and let $\tau_0 > 0$ satisfy

$$S(\tau_0) := \frac{E - cB}{3} - \frac{8}{35}\, cL\tau_0^{1/2} - \frac{128}{3465}\, L\tau_0^{3/2} - \frac{B}{15}\, \tau_0 > 0. \tag{21}$$

Then $K(\tau) \geq \frac{1}{3}\, \langle u^3 \rangle \tau^3 + S(\tau_0)\tau^4 > 0$ on $(0, \tau_0]$.

Proof. By (7) and Proposition 19, integrate against $2(\tau - a - b) \geq 0$ on the triangle. Dirichlet integrals give the weights $2 \iint (\tau - a - b) = \tau^3/3$, $2 \iint (\tau - a - b)(a + b) = \tau^4/6$, $2 \iint (\tau - a - b)4ab = \tau^5/15$, $2 \iint (\tau - a - b)a^{3/2} = \frac{16}{315}\, \tau^{9/2}$, $2 \iint (\tau - a - b)(a + b)^{3/2} = \frac{8}{63}\, \tau^{9/2}$, $2 \iint (\tau - a - b)ba^{3/2} = \frac{32}{3465}\, \tau^{11/2}$. Thus

$$K \geq \tfrac{1}{3}\, \langle u^3 \rangle \tau^3 + \tfrac{1}{3}\, (E - cB)\tau^4 + \tfrac{c}{15}\, \tau^5 - \tfrac{8}{35}\, cL\tau^{9/2} - \tfrac{128}{3465}\, L\tau^{11/2} - \tfrac{B}{15}\, \tau^5,$$

and $S(\tau)$ decreases in τ. □

Remark 22. So $K = \frac{1}{3}\, \langle u^3 \rangle \tau^3 + \frac{1}{3}\, E\tau^4 + O(\tau^{9/2})$. For the triangle (inradius 1), $E = 9/100 - 27/700 = 9/175$, which reproduces the coefficient in equation (14) of the source, $\mu_3 \sim \frac{27}{19250}\, \mathrm{Pe}^3 \tau^3 + \frac{9}{175}\, \mathrm{Pe}^3 \tau^4$. For the disk $E = 1/24$, so $K \sim \tau^4/72$. The point of Proposition 20 is that E stays near $1/24$ as $n \to \infty$ while $\langle u^3 \rangle \to 0$: the window $(0, \tau_0]$ does not shrink. ⌟

For a polynomial $\tilde{u}_L$ with $-\Delta\tilde{u}_L = 2$ on P_n let $\theta_- \le \tilde{u}_L \le \theta_+$ on ∂P_n and let $\bar{m}$ bound $|\nabla\tilde{u}_L|$ on ∂P_n; then $|\nabla\tilde{u}_L| \le \bar{m}$ on P_n, as $|\nabla\tilde{u}_L|^2$ is subharmonic. The maximum principle gives $\tilde{u} \le \tilde{u}_L - \theta_-$ and $\tilde{u}_L \le \theta_+ + \bar{m}d$, so Lemma 18 applies with $\beta = \theta_+ - \theta_-$. Also c and $\|u\|$ are enclosed by the corresponding quantities of $\tilde{u}_L$, within $\theta_+ - \theta_-$.

4 The third moment

Proposition 23. $\langle u^3\rangle > 0$ for every $n \ge 3$.

For $n = 3$ the value is $27/19250$ (Section 7). For $n \ge 4$ we use the following.

Lemma 24. On P_n let $Y = \frac{1}{2}(1 - r^2) + t^2/6$ and $H = \tilde{u} - Y$. Then H is harmonic, $H = \sigma^2/2 - t^2/6$ on the side at arclength σ from its midpoint, $-t^2/6 \le H \le t^2/3$, and $H_c = H - \langle H\rangle$ satisfies $\|H_c\|^2 \le \frac{t^4}{16}\left(\frac{1}{n+1} + \frac{4}{3}t^2\right)$.

Proof. On the unit circle, inside P_n, $|H - t^2/12| \le t^2/4$ by the maximum principle. There $H = \beta_0 + \sum_{k\ge1}\beta_k r^{nk}\cos(nk\theta)$, so $\frac{1}{2}\sum\beta_k^2 \le t^4/16$, and the circular mean of $(H-\beta_0)^2$ at radius $r \le 1$ is at most $r^{2n}t^4/16$. Integrating, $\int_{r<1}(H-\beta_0)^2 \le \frac{\pi}{n+1}\frac{t^4}{16}$. On $P_n\backslash\{r<1\}$, of area $nt - \pi \le nt\cdot t^2/3$, $|H - \beta_0| \le t^2/2$. Divide by $|P_n| = nt \ge \pi$, and note $\|H_c\| \le \|H - \beta_0\|$. □

Let $Y_c = Y - \langle Y\rangle = \frac{1}{4} + \frac{t^2}{12} - \frac{r^2}{2}$, $\phi = Y_c^2 - \langle Y_c^2\rangle$, Φ the radial polynomial with $-\Delta\Phi = \phi$, and $\varrho_\Phi = \Phi -$ (mean of Φ on the side) on ∂P_n. Since $u = Y_c + H_c$, Green's identity gives

$$\langle u^3\rangle = M(t) + \frac{3}{|P_n|}\oint \varrho_\Phi\,\partial_n H + 3\langle Y_c H_c^2\rangle + \langle H_c^3\rangle, \qquad M(t) = \langle Y_c^3\rangle - \frac{3}{|P_n|}\oint H\,\partial_n\Phi = \frac{t^4}{360} + \frac{19t^6}{7560},$$

the last equality being an exact polynomial identity. By Lemma 11, $|\frac{3}{|P_n|}\oint\varrho_\Phi\partial_n H| \le 6(t+R)t^5(q(t)/3)^{1/2}$, where $\int_0^t \varrho_\Phi^2 = t^9 q(t)$, $q(0) = 1/4082400$. By Lemma 24 and $1/(n+1) \le t/\pi$, $|3\langle Y_c H_c^2\rangle + \langle H_c^3\rangle| \le (\frac{3}{4} + \frac{5}{4}t^2 + \frac{1}{2}t^2)\,\frac{t^4}{16}\,(\frac{t}{\pi} + \frac{4}{3}t^2)$. In interval arithmetic the resulting lower bound for $\langle u^3\rangle/t^4$ is at least $1.06\cdot10^{-3}$ on $(0, \tan(\pi/40)]$ (`u3_uniform.py`), so $\langle u^3\rangle > 0$ for $n \ge 40$, and $\langle u^3\rangle = t^4/360 + O(t^{9/2})$. The sharper remainder $O(t^5)$ of Theorem 4 uses the explicit Fourier approximant of [3]; `sg_uniform.py` there gives an independent proof for $n \ge 140$.

For $4 \le n \le 150$, let $\tilde{u}_N$ be a polynomial with $-\Delta\tilde{u}_N = 2$, $E = \tilde{u} - \tilde{u}_N$, $f = E - \langle E\rangle$, $u_N = \tilde{u}_N - \langle\tilde{u}_N\rangle$, and Φ a polynomial with $-\Delta\Phi = u_N^2 - \langle u_N^2\rangle$. Then $\langle u^3\rangle = \langle u_N^3\rangle + 3|P_n|^{-1}\oint(\Phi\partial_n E + \tilde{u}_N\partial_n\Phi) + 3\langle u_N f^2\rangle + \langle f^3\rangle$, and Lemma 11 bounds the last terms and the term containing $\partial_n E$. The script `sg_certify.py` of [3] evaluates this in ball arithmetic; we re-ran it for this paper, and all 147 cases pass.

5 Certified computation: $4 \le n \le 199$

Fix $\tau_0 = 0.015$. For each n the script `finite_n.py` checks (21) and $K > 0$ on $[\tau_0, \infty)$.

Exact integration. Let $p = x^2 + y^2$, $q = \mathrm{Re}(x + \mathrm{i}y)^n$. Every D_n-invariant polynomial is a polynomial in p, q. For a homogeneous function F_d of degree d, cone coordinates on T give $\langle F_d\rangle = \frac{2}{d+2}\frac{1}{t}\int_0^t F_d(1,\sigma)\,\mathrm{d}\sigma$. The integrand is a polynomial in σ, integrated exactly by Gauss–Legendre rules with ball enclosures of nodes and weights. Thus all

n	M
4	39

Table 26 / Certificates of Proposition 25, sample of n. All numbers are outward rounded to the digits shown, $\tau_0 = 0.015$. Full data: `certificates/finite_4_200.json`.

moments $\langle p^a q^k\rangle$ are enclosed, and products reduce to these moments. Also $\Delta(p^a q^k) = 4a(a+nk)p^{a-1}q^k + n^2k(k-1)p^{a+n-1}q^{k-2}$, and on the side $\partial_x p = 2$, $\partial_x q = n\,\mathrm{Re}(1+\mathrm{i}\sigma)^{n-1}$. All arithmetic is in Arb [8] through python-flint, at 3000 or 5000 bits.

Approximants. With $h_k = \mathrm{Re}((x+\mathrm{i}y)/R)^{nk}$ we take $\tilde u_H = \frac12(1-p) + \sum_{k=0}^{K_H} a_k h_k$ and $g_H = \frac{p^2}{32} + \beta p + \sum_{k=1}^{K_H}(\gamma_k p + b_k)h_k$, with $\gamma_k = -a_k/(4(1+nk))$, $\beta = -\frac14\,(\frac12 - \langle\tilde u_H\rangle + a_0)$, so that $-\Delta g_H = \tilde u_H - \langle\tilde u_H\rangle$ exactly. The coefficients are dyadic numbers from floating-point least squares on the side. Lemma 12 involves only one-dimensional integrals, so $K_H = 24$ to 40 is cheap. A second pair $(\tilde u_L, g_L)$ of the same form, with $K_L \le 8$, enters all two-dimensional integrals; $\|g_H - g_L\|$ is computed from the Gram matrix of the functions p, ph_k, h_k by the cone formula. So $\omega = \|g - g_L\|$ is enclosed, with $\hat g = g_L - \langle g_L\rangle$. We take $w = \tilde u_L - \langle\tilde u_L\rangle$ in Lemma 14; then $o = \theta_+ - \theta_-$, and U comes from a grid of T plus the Lipschitz bound.

Modes. On the span of the scaled monomials $(p/R^2)^a(q/R^n)^k$ with $(k=0,\ a\le A_0)$ or $(1\le k\le K_q,\ a\le A_1)$ we compute Gram and stiffness matrices in ball arithmetic, a Cholesky factor, and the Ritz pairs in double precision. We keep the M pairs with $\mu_k \le 800$ and subtract means; μ_k, d_k and the coefficients are exact binary numbers. How they were found is irrelevant for rigour. Then $\|\ell_k\|_* \le C_P\|R_k - \langle R_k\rangle\| + \tau_{\mathrm{tr}}\|\partial_n\psi_k\|_\partial$ with $R_k = -\Delta\psi_k - \mu_k\psi_k$, and ρ, L_1, Δ_0 of Lemma 13 and the matrix W_{kl} of (15) are enclosed.

The inequality. On the grid $\tau_{j+1} = 1.02\,\tau_j$ (for $\tau_j < 0.2$), $\tau_{j+1} = 1.1\,\tau_j$ (beyond), from τ_0 to 40, the script enclosures $\tilde K, \tilde K', \tilde K''$ at τ_j and verifies, with the bound (16) on $\tilde K'''$ and $\mathcal{E}$ increasing, that the lower Taylor bound on $[\tau_j, \tau_{j+1}]$ exceeds $\mathcal{E}(\tau_{j+1})$. For $\tau \ge 40$ it verifies $(2W_{00} - e_1)40 > C_1$, with $e_1\tau \ge \mathcal{E}(\tau)$ the linear bound of Lemma 14. In particular $2W_{00} - e_1 > 0$ gives $\langle ug^2\rangle > 0$.

Proposition 25. For every $4 \le n \le 200$, $K > 0$ on $[0.015, \infty)$ and $S(0.015) > 0$.

Table 26 shows sample data. The smallest value of (lower bound $-\,\mathcal{E}$)$/\tilde K$ over all n and grid points is attained at τ_0; the table gives it, with $\tilde K(\tau_0)$ and $\mathcal{E}$ at the next grid point.

6 All $n \ge 200$ and the disk

Now rescale P_n to area π: call it $\mathcal{P}$, with apothem a, $a^2 = \arctan(t)/t$, and circumradius $v = aR$. The alternating series of arctan gives $1 - t^2/3 \le a^2 \le 1 - t^2/3 + t^4/5$ and $v^2 \le 1 + \frac23 t^2$. Time scales by a^2. Let j_k be positive zeros of J_1, $\mu_k = j_k^2$, $\psi_k(r) = J_0(j_k r)/J_0(j_k)$, and on the unit disk D

$$\bar u = \tfrac14 - \tfrac12 r^2, \qquad g_0 = \tfrac{1}{32} r^4 - \tfrac{1}{16} r^2 + \tfrac{1}{48}, \qquad -\Delta g_0 = \bar u, \quad g_0'(1) = 0, \quad \langle g_0\rangle_D = 0.$$

Lemma 27 (disk data)**.** In $L^2(D)$ with the average product, the ψ_k are orthonormal, $\langle\bar{u}\psi_k\rangle_D = -2/\mu_k$, $\langle g_0\psi_k\rangle_D = -2/\mu_k^2$, $\langle g_0^2\rangle_D = 1/11520$, $\langle\bar{u}g_0^2\rangle_D = 1/184320$, and

$$\langle\bar{u}\psi_k^2\rangle_D = \frac{1}{12}\,, \quad \langle\bar{u}\psi_k\psi_l\rangle_D = -\frac{2(\mu_k+\mu_l)}{(\mu_k-\mu_l)^2}\ (k\neq l), \quad \langle\bar{u}g_0\psi_k\rangle_D = \frac{\mu_k^2+96\mu_k-1728}{48\mu_k^3}\,.$$

Proof. For a radial polynomial f, Green's identity with $\psi_k(1)=1$, $\psi_k'(1)=0$ gives $\langle f\psi_k\rangle_D = \mu_k^{-1}(2f'(1) - \langle\psi_k\Delta f\rangle_D)$, which terminates. For the products, let $E = \int_0^1 r^2\psi_l\psi_k'\,\mathrm{d}r$, $F = \int_0^1 r^2\psi_k\psi_l'\,\mathrm{d}r$ with $k\neq l$. Integration by parts gives $E+F=1$. Multiplying the two Bessel equations by $r^2\psi_l'$, $r^2\psi_k'$ and adding gives $\mu_k F + \mu_l E = 0$. Also $(\mu_k-\mu_l)2\int_0^1 r^3\psi_k\psi_l\,\mathrm{d}r = 4(E-F)$. So $\langle r^2\psi_k\psi_l\rangle_D = 4(\mu_k+\mu_l)/(\mu_k-\mu_l)^2$. The Pohozaev identity $\int_0^1 r^4\psi_k'(\psi_k''+\psi_k'/r+\mu_k\psi_k)\,\mathrm{d}r = 0$ gives $\langle r^2\psi_k^2\rangle_D = 1/3$. □

With $D_0=1$, $\psi_0=g_0$, $\mu_0=0$, $D_k=2/\mu_k^2$, $M=12$, let $W_{kl}=D_kD_l\langle\bar{u}\psi_k\psi_l\rangle_D$ and define K_D by (15). This is a fixed function, independent of n. On $\mathcal{P}$ we use $\hat{g}=g_0-\langle g_0\rangle$ and $\psi_k-\langle\psi_k\rangle$, with averages over $\mathcal{P}$. Set $x=t^2$.

Lemma 28. On $\mathcal{P}$, let $C_P = v/\pi$, $\tau_{\mathrm{tr}}^2 = (2C_P^2+2vC_P)/a$. The following bounds hold, every right-hand side being increasing in t:

(1) $H=\tilde{u}-\frac{1}{2}(1-r^2)$ is harmonic, its oscillation on $\mathcal{P}$ is at most $x/2$, and $\|H_c\| \le \frac{x}{4}(t/\pi+\frac{4}{3}x)^{1/2}$;

(2) the flux of g_0 is $e_0=\frac{a}{8}(r^2-1)$ on the side, and $\|e_0\|_\partial^2 \le x^2Q/32$, where $Q=\frac{A^2}{5}+\frac{2}{3}A(\zeta-\frac{1}{3})+(\zeta-\frac{1}{3})^2$, $A=a^2=1-\frac{x}{3}+x\zeta$, $0\le\zeta\le x/5$;

(3) $\omega := \|g-\hat{g}\| \le C_P(C_P\|H_c\|+\tau_{\mathrm{tr}}\|e_0\|_\partial)$;

(4) the flux of ψ_k satisfies $|\partial_n\psi_k| \le 16\,\Psi_k|e_0|$, where $\Psi_k=\sup|\Psi''|$ over $[a^2,v^2]$, $\Psi(r^2)=\psi_k(r)$; so $\|\ell_k\|_* \le 16\tau_{\mathrm{tr}}\Psi_k\|e_0\|_\partial$;

(5) with $\zeta_M = g_0-\sum_{k\le M}d_k\psi_k$, $d_k=-2/\mu_k^2$: $\rho^2 \le \frac{1}{11520}-\sum_{k\le M}d_k^2+\frac{x}{3}\sup_{[1,v]}\zeta_M^2$;

(6) for radial Lipschitz f: $|\langle f\rangle_{\mathcal{P}}-\langle f\rangle_D| \le w_1\,\mathrm{Lip}\,f$, $w_1=x^2/6$; and $\langle\bar{u}\rangle_{\mathcal{P}}=-\bar{b}/2$, $0\le\bar{b}=\frac{1}{2}(a^2(1+\frac{x}{3})-1)\le x^2/20$.

Proof. (1) As in Lemma 24, on $\mathcal{P}$, with inscribed circle of radius a. The wall values of H are $\frac{1}{2}(r^2-1)$. (2) $g_0'=r(r^2-1)/8$, and on the side $\partial_x=(a/r)\partial_r$; then integrate. (3) Lemma 12's proof: here $u-(\bar{u}-\langle\bar{u}\rangle)=H_c$ and $\langle\nabla(g-\hat{g})\cdot\nabla\varphi\rangle = \langle H_c\varphi\rangle - |\mathcal{P}|^{-1}\oint e_0\varphi$. (4) $\partial_n\psi_k = 2a\Psi'(r^2)$ and $\Psi'(1)=0$, so $|\partial_n\psi_k| \le 2a\Psi_k|r^2-1| = 16\Psi_k|e_0|$; ψ_k solves the Helmholtz equation exactly, so ℓ_k is only the boundary term. (5) $\langle\zeta_M^2\rangle_D = \langle g_0^2\rangle_D - \sum d_k^2$, by orthonormality; and $|\mathcal{P}\backslash D| \le \pi(1-a^2) \le \pi x/3$. (6) The laws of r under $\mathcal{P}$ and D agree on $[0,a]$ and their distribution functions differ by at most $1-a^2\le x/3$ on $[a,v]$, with $v-a\le x/2$. □

Proposition 29. For $N=200$, every $n\ge N$, and $\hat{\tau}_0=0.0149$: $K_n>0$ on $[\hat{\tau}_0,\infty)$ in the time unit of $\mathcal{P}$, and $S(0.015)>0$ in the apothem-1 frame. The same holds for

the disk.

Proof. By Lemmas 13 and 14 with $w = \bar{u} - \langle \bar{u} \rangle_{\mathcal{P}}$, $U = \frac{1}{4} + \frac{x}{3} + \frac{\bar{b}}{2}$, $o = x/2$, and by Lemma 28(6), which replaces averages over $\mathcal{P}$ by averages over D,

$$K_n(\tau) \geq K_D(\tau) - \mathcal{E}(\tau) - r_W \tau, \qquad r_W = 2\big(w_1(vF^2 + 4UFL) + \tfrac{1}{2}\bar{b}F^2\big),$$

where $F \geq \sup|f_s|$, $L \geq \sup|f_s'|$ on $[0, v]$ for $f_s = g_0 - \sum_k d_k \mathrm{e}^{-\mu_k s}\psi_k$; we use $|J_0| \leq 1$, $|J_1| \leq 1/\sqrt{2}$. For the term r_W, note $\langle w\tilde{A}_a\tilde{A}_b\rangle_{\mathcal{P}}$ differs from $\langle \bar{u} f_a f_b\rangle_D$ by at most $w_1 \operatorname{Lip}(\bar{u} f_a f_b) + \frac{\bar{b}}{2} F^2 + 2w_1 LUF$. All constants are increasing in t; `uniform.py` evaluates them at $t = \tan(\pi/200)$: $\omega \leq 4.2 \cdot 10^{-6}$, $L_1 \leq 1.6 \cdot 10^{-6}$, $\rho \leq 1.9 \cdot 10^{-6}$, and checks the inequality on the grid of Section 5 and for $\tau \geq 40$. At $\hat{\tau}_0$: $K_D = 3.06 \cdot 10^{-10}$, error $2.06 \cdot 10^{-10}$. For the disk, $t = 0$, all errors vanish except the truncation $\rho \leq 1.1 \cdot 10^{-6}$. In the apothem-1 frame, for S: $H' = \tilde{u} - Y$ has $-x/6 \leq H' \leq x/3$, so $c \in [\frac{1}{4} - \frac{x}{12}, \frac{1}{4} + \frac{5x}{12}]$, $\|\tilde{u}\| \leq \|Y\| + x/3$, $E = 2c^2 - \|\tilde{u}\|^2$, and $\tilde{u} \leq Rd + x/2$ because $Y \leq x/6 + Rd$; so $\bar{m} = R$, $\beta = x/2$. □

Proof (Proof of Theorem 2). Let $n \geq 4$. $\langle u^3 \rangle > 0$ by Proposition 23. By Proposition 20 with Proposition 25 ($n \leq 199$) or Proposition 29 ($n \geq 200$), $K > 0$ on $(0, 0.015]$ in the apothem-1 frame. On $[0.015, \infty)$ use Proposition 25; for $n \geq 200$, Proposition 29 covers $\tau \geq \hat{\tau}_0/a^2$, and $\hat{\tau}_0/a^2 \leq 0.015$ since $a^2 \geq 1 - t^2/3$. The disk is the same with $\langle u^3 \rangle = 0$, $\bar{m} = 1$, $\beta = 0$, $c = \frac{1}{4}$, $E = \frac{1}{24}$. □

7 The equilateral triangle

For $n = 3$, $R = 2$, and everything except the modes is an exact polynomial in p and $q = \operatorname{Re}(x+\mathrm{i}y)^3$: $\tilde{u} = \frac{1}{2}(1-p) + \frac{1}{6}(1-q)$, $c = \frac{3}{10}$, $\langle u^2 \rangle = \frac{27}{700}$, $\langle u^3 \rangle = \frac{27}{19250}$, $-\frac{3}{10} \leq u \leq \frac{11}{30}$, and

$$g = -\frac{p^3}{960} + \frac{p^2}{32} - \frac{11p}{120} + \frac{pq}{96} + \frac{q^2}{480} + \text{const}, \qquad \langle ug \rangle = \frac{9}{1540}, \quad \langle ug^2 \rangle = -\frac{7101}{476476000}.$$

So $\omega = o = 0$. The long-time coefficient in the normalisation of the source is $3\langle ug_2 \rangle / (2\langle ug \rangle)^{3/2} = -789\sqrt{770}/309400 = -0.0707623\ldots$, and the geometric skewness is $\langle u^3 \rangle / \langle u^2 \rangle^{3/2} = 4\sqrt{21}/99$.

For derivatives: with $e = A - \tilde{A}$ one has $K' - \tilde{K}' = 2\int_0^\tau \langle ue(s)(P_{\tau-s}u + \tilde{A}'(\tau - s))\rangle \, \mathrm{d}s + 2\langle u\tilde{A}(0)e(\tau)\rangle$, by differentiating the identity of Lemma 14. So, by (9),

$$|K' - \tilde{K}'| \leq 2U\Delta_0\Big(\frac{\|u\|}{\nu} + \sum_k |d_k|\|\psi_k\| + \rho\Big). \tag{30}$$

Proof (Proof of Theorem 3). Proposition 20 with $\bar{m} = \frac{3}{2}$, $\beta = 0$, $L = 8 \cdot \frac{3}{2}/(3\sqrt{\pi})$: the function $h(x) = \frac{1}{3}\langle u^3 \rangle + \frac{E}{3}x^2 - \frac{8}{35}cLx^3 - \frac{128}{3465}Lx^5$, $x = \sqrt{\tau}$, increases and then decreases, and $h(0), h(\sqrt{0.025}) > 0$; so $K > 0$ on $(0, 0.025]$. With 35 Ritz modes ($\mu_k \leq 400$, $\rho + L_1 \leq 1.5 \cdot 10^{-5}$), `triangle.py` verifies $K > 0$ on $[0.025, 1]$, $K' < 0$ on $[1, 1.6]$ by (30), $K < 0$ on $[1.6, \infty)$, and locates the zero. □

8 Table 1 of the source

Table 31 compares the printed values (arXiv v3; circumradius 1) with enclosures from [3], script `table1.py`, re-run for this paper. All printed values agree up to rounding, except

n	long-time coefficient, source	enclosure	S^G, source	enclosure
3	-0.035	$-789\sqrt{770}/618800 = -0.0353812$	0.185	$4\sqrt{21}/99 = 0.1851546$
4	0.139	$[0.13849, 0.13856]$	0.0813	$[0.081294, 0.081297]$
5	0.194	$[0.19390, 0.19429]$	0.042	$[0.042262, 0.042270]$
6	0.215	$[0.21474, 0.21603]$	0.024	$[0.024264, 0.024274]$
7	0.225	$[0.22384, 0.22699]$	0.0149	$[0.014939, 0.014955]$
8	0.231	$[0.22903, 0.23281]$	0.0097	$[0.0097036, 0.0097168]$
20	0.243	$[0.23937, 0.24655]$	$3.97 \cdot 10^{-4}$	$[3.9611, 3.9832] \cdot 10^{-4}$
100	0.245	$[0.24384, 0.24589]$	$8.6 \cdot 10^{-7}$	$[8.3475, 8.3971] \cdot 10^{-7}$
∞	0.245	$\sqrt{6}/10 = 0.2449490$	0	0

Table 31 / Long-time coefficient $3\langle ug_2\rangle/(2\langle ug\rangle)^{3/2}$ and geometric skewness $S^G = \langle u^3\rangle/\langle u^2\rangle^{3/2}$, circumradius 1.

the arXiv-only $S^G_{100} = 8.6 \cdot 10^{-7}$, about 3% too large; it does not appear in the published table.

9 Computer verification and limitations

All scripts are in `all-times/scripts/` unless noted, print PASS or FAIL per claim, and exit non-zero on failure. Environment: Python 3.12, python-flint 0.9.0 (Arb), numpy, scipy, sympy, under WSL Fedora. `run_all.sh` runs everything; logs are in `all-times/logs/`.

script	claim	time
`run_finite.py 4 200`	Proposition 25, using `finite_n.py`, `modal_lib.py`	1 h on 3 co
`uniform.py 200`, `uniform.py 0`	Proposition 29, large n and the disk	10 s
`u3_uniform.py 40`	$\langle u^3\rangle > 0$, $n \geq 40$	5 s
`../scripts/sg_certify.py 4 150`	$\langle u^3\rangle > 0$, $4 \leq n \leq 150$	15 min
`../scripts/sg_uniform.py 140 6`	$\langle u^3\rangle > 0$, $n \geq 140$, second proof	5 s
`triangle.py`	Theorem 3	2 min
`../scripts/triangle_exact.py`, `table1.py`	exact triangle values, Table 31	5 min
`numerics_check.py`	floating-point cross-checks, not a certificate	3 min

Cross-checks. For the square, K from the exact lattice sum $G(a,b) = \sum_{\xi,\eta\in\mathbb{Z}^2} \hat{u}_\xi \hat{u}_\eta \hat{u}_{\xi+\eta} \mathrm{e}^{-\pi^2(|\xi|^2 a + |\eta|^2 b)}$ agrees with the Galerkin value to four digits at $\tau = 0.015$ ($8.20 \cdot 10^{-10}$), and with the certified $\tilde{K}$. The law in Remark 22 is confirmed numerically and against equation (14) of the source.

Trusted base. Payne–Weinberger constant [5] [6]; H^2 regularity on convex polygons [7]; the spectral theorem for the Neumann Laplacian; Green's identities; Arb and python-flint for ball arithmetic; sympy for exact rational identities. Floating point only chooses coefficients.

Limitations. The proof is computer-assisted and has not been verified independently. The window $\tau_0 = 0.015$ and the threshold $N = 200$ are technical. Numerics suggest more than we prove: $G(a,b) > 0$ on the whole quadrant for every $n \geq 4$, so that K is

convex (and $\langle u e_1^2 \rangle > 0$ for the first invariant eigenfunction, already in the square, where it is about $+0.023$); we did not prove this. The monotonicity in n suggested by Table 1 of the source is not addressed, nor are the unimodality and median statements of the source.

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